

Introduction to Probability Theory

a) Basics of Probability Probability measures how likely an event is, on a scale from 0 (impossible) to 1 (certain).

- Sample space (S): all possible outcomes of an experiment
- Event (E): a subset of outcomes you're interested in
- Formula (equally likely outcomes): $P(E) = \text{favorable outcomes} / \text{total outcomes}$
- Addition rule: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
- If A and B are mutually exclusive: $P(A \text{ or } B) = P(A) + P(B)$
- Independent events: $P(A \text{ and } B) = P(A) \times P(B)$

Example: Rolling a die, $P(\text{even number}) = 3/6 = 1/2$

b) Conditional Probability The probability of event A occurring, given that event B has already happened. It "updates" your probability using new information.

- Formula: $P(A|B) = P(A \text{ and } B) / P(B)$, where $P(B) > 0$
- Multiplication rule: $P(A \text{ and } B) = P(A|B) \times P(B) = P(B|A) \times P(A)$
- If independent: $P(A|B) = P(A)$ — knowing B tells you nothing about A

Example: From a deck of cards, $P(\text{King} | \text{Face card}) = P(\text{King and Face})/P(\text{Face}) = (4/52)/(12/52) = 1/3$

c) Bayes' Theorem Lets you reverse a conditional probability — go from $P(\text{evidence}|\text{cause})$ to $P(\text{cause}|\text{evidence})$. Useful for reasoning backward from evidence to cause.

- Formula: $P(A|B) = [P(B|A) \times P(A)] / P(B)$
- $P(A)$ = prior (belief before evidence)
- $P(B|A)$ = likelihood (chance of evidence if A is true)
- $P(A|B)$ = posterior (updated belief after evidence)
- $P(B)$ can be expanded using total probability: $P(B) = P(B|A)P(A) + P(B|\text{not } A)P(\text{not } A)$

Example: A disease affects 1% of people. Test is 99% accurate. $P(\text{Disease} | \text{positive test}) = (0.99 \times 0.01) / [(0.99 \times 0.01) + (0.01 \times 0.99)] = 50\%$ Even with a highly accurate test, a positive result is only 50% likely to be correct — because the disease itself is rare (this is the "base rate" effect).

d) Random Variables and Probability Distributions A random variable (RV) assigns a numerical value to each outcome of a random experiment, so we can do math on outcomes.

Discrete RV — takes countable values, described by a Probability Mass Function (PMF): $P(X = x)$

- Bernoulli: single trial, success/failure
- Binomial: number of successes in n trials
- Poisson: number of events in a fixed interval

Continuous RV — takes any value in a range, described by a Probability Density Function (PDF), where probability = area under the curve

- Uniform: equally likely across a range
- Normal: the bell curve, defined by mean (μ) and standard deviation (σ)
- Exponential: waiting time until an event

Key summary measures (apply to any RV):

- Expected value (mean): $E[X]$ — long-run average
- Variance: $\text{Var}(X)$ — spread around the mean
- Standard deviation: $\sqrt{\text{Var}(X)}$ — spread in original units
- CDF: $F(x) = P(X \leq x)$ — cumulative probability up to x

a) Population vs. Sample

- Population: the entire group you want to study or draw conclusions about (e.g., all voters in a country)
- Sample: a subset of the population actually collected and measured (e.g., 1,000 surveyed voters)
- Parameter: a numerical value describing the population (e.g., population mean μ , population proportion p) — usually unknown
- Statistic: a numerical value describing the sample (e.g., sample mean $\bar{x}$, sample proportion $\hat{p}$) — calculated from data, used to estimate the parameter
- Why sample instead of studying the whole population: cost, time, and practicality — sometimes measuring the whole population is impossible (e.g., destructive testing) or too expensive

b) Sampling Techniques Probability sampling (every unit has a known chance of selection — allows valid statistical inference):

- Simple random sampling: every member has an equal chance of being chosen (e.g., drawing names from a hat)
- Systematic sampling: select every k th member from a list (e.g., every 10th person)
- Stratified sampling: divide population into subgroups (strata) by a characteristic (age, gender, etc.), then randomly sample from each stratum — ensures representation of subgroups
- Cluster sampling: divide population into clusters (e.g., geographic regions), randomly select whole clusters, then sample everyone (or randomly) within chosen clusters — cheaper for spread-out populations

Non-probability sampling (no random selection — faster/cheaper but can introduce bias):

- Convenience sampling: whoever is easiest to reach (e.g., surveying people at a mall)
- Quota sampling: fill predetermined quotas for subgroups without random selection
- Judgmental/purposive sampling: researcher handpicks based on judgment

c) Point Estimation and Interval Estimation

- Point estimation: a single value used to estimate a population parameter
 - Example: sample mean $\bar{x}$ estimates population mean μ
 - Common estimators: sample mean (estimates μ), sample proportion (estimates p), sample variance (estimates σ^2)
 - Limitation: gives no idea of how much uncertainty/error is involved
- Interval estimation: a range of values, instead of a single number, likely to contain the true parameter
 - Comes with a confidence level (e.g., 95%) indicating how reliable the interval is
 - More informative than a point estimate because it communicates precision/uncertainty

Example: Point estimate — average height = 170 cm. Interval estimate — average height is between 168 cm and 172 cm, with 95% confidence.

d) Confidence Intervals A confidence interval (CI) is a range of values, calculated from sample data, likely to contain the true population parameter.

General formula: $CI = \text{point estimate} \pm \text{margin of error}$

For population mean (large sample, known σ): $CI = \bar{x} \pm z \times (\sigma / \sqrt{n})$

- $\bar{x}$ = sample mean
- z = z-score for desired confidence level (1.96 for 95%, 2.58 for 99%, 1.645 for 90%)
- σ = population standard deviation
- n = sample size

If σ is unknown (common in practice), use the sample standard deviation (s) and t-distribution instead of z : $CI = \bar{x} \pm t \times (s / \sqrt{n})$

Interpretation: A 95% confidence interval means that if you repeated the sampling process many times, about 95% of the calculated intervals would contain the true population parameter. (It does NOT mean "95% probability the true value is in this specific interval.")

Key relationship: Larger sample size $\rightarrow$ smaller margin of error $\rightarrow$ narrower, more precise interval. Higher confidence level $\rightarrow$ wider interval (more certainty needs more room).

Example: A sample of 100 students has mean test score 75, $s = 10$. $95\% CI = 75 \pm 1.96 \times (10/\sqrt{100}) = 75 \pm 1.96 = (73.04, 76.96)$

What is Hypothesis Testing?

Suppose a company claims:

"Our new fertilizer increases crop yield."

Can we simply believe them?

No.

We collect data from farms and use statistics to determine whether there is enough evidence to support the claim.

This process is called **hypothesis testing**.

A hypothesis test doesn't prove something is true; it tells us whether the evidence is strong enough to support a claim.

a) Null Hypothesis (H_0) and Alternative Hypothesis (H_1 or H_a)

A **hypothesis** is simply an assumption or claim about a population.

Every hypothesis test starts with **two competing hypotheses**.

1. Null Hypothesis (H_0)

The **Null Hypothesis** is the default assumption.

It usually states:

- Nothing has changed.
- There is no relationship.
- There is no difference.
- No effect exists.

Think of it as **"innocent until proven guilty."**

Examples

A medicine company says a new drug works.

The null hypothesis is:

The drug has **no effect**.

Symbolically:

$$H_0 : \mu = \mu_0$$

where

- μ = actual population mean
- μ_0 = claimed mean

Another example

A teacher says the average score is 75.

$$H_0 : \mu = 75$$

Meaning:

"The average score is still 75."

2. Alternative Hypothesis (H_1)

The **Alternative Hypothesis** is what the researcher wants to prove.

It states that

- Something changed.
- A relationship exists.
- An effect exists.

- The population differs from the claim.

Examples

Drug example

H_1 : The drug improves recovery.

Exam example

$H_1 : \mu > 75$

Meaning:

"The average score is greater than 75."

Simple Analogy

Imagine a courtroom.

- **Null Hypothesis:** The accused is innocent.
- **Alternative Hypothesis:** The accused is guilty.

The court assumes innocence until enough evidence proves guilt.

Statistics works exactly the same way.

Types of Alternative Hypotheses

Right-tailed Test

Looking for an increase.

$H_1 : \mu > \mu_0$

Example

The new machine produces **more** products.

Left-tailed Test

Looking for a decrease.

$$H_1 : \mu < \mu_0$$

Example

The new medicine lowers blood pressure.

Two-tailed Test

Looking for any difference.

$$H_1 : \mu \neq \mu_0$$

Example

The average weight has changed.

b) Type I Error and Type II Error

Even after collecting data, we can make mistakes.

There are two possible mistakes.

Type I Error

Rejecting the Null Hypothesis even though it is actually true.

In simple words,

You think something changed, but it didn't.

This is called a **False Positive**.

Example

Suppose

The medicine actually does **not** work.

But your experiment concludes

"The medicine works."

That is a Type I Error.

Court Analogy

An innocent person is declared guilty.

Machine Learning Example

Spam Filter

A genuine email is classified as spam.

The email is actually normal, but the model wrongly flags it.

False Positive.

Type II Error

Failing to reject the Null Hypothesis even though it is false.

You think nothing changed, but it actually did.

This is called a **False Negative**.

Example

The medicine actually works.

Your experiment concludes

"The medicine doesn't work."

That is a Type II Error.

Court Analogy

A guilty person is declared innocent.

Machine Learning Example

A spam email is classified as a normal email.

The model misses the spam.

False Negative.

Comparison

Type I Error

False Positive

Reject H_0 incorrectly

Think there is an effect when there isn't

Type II Error

False Negative

Accept H_0 incorrectly

Think there is no effect when there is

c) Significance Level (α)

The **significance level**, denoted by α (**alpha**), is the maximum probability of making a **Type I Error** that we are willing to accept.

In other words,

"How much risk are we willing to take of rejecting a true null hypothesis?"

Common choices are:

- 0.05 (5%)
- 0.01 (1%)
- 0.10 (10%)

The most common is:

$$\alpha = 0.05$$

This means:

"We accept a 5% chance of making a Type I Error."

Why 5%?

There is nothing magical about 5%.

It is simply a widely accepted balance between being too strict and too lenient.

- Smaller α (0.01): Harder to reject H_0 , fewer false positives.
 - Larger α (0.10): Easier to reject H_0 , more false positives.
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p-value

The **p-value** is one of the most misunderstood concepts in statistics.

It is **not** the probability that the null hypothesis is true.

Instead, it answers this question:

If the null hypothesis were actually true, how likely is it to observe data this extreme (or even more extreme)?

Think of it as a measure of how surprising your data is under the assumption that (H_0) is true.

Example

Suppose a company claims its bulbs last **1000 hours**.

You test several bulbs and get an average of **950 hours**.

If this difference is very unlikely when the true mean is 1000 hours, the p-value will be very small.

Decision Rule

- If

$$p \leq \alpha$$

Reject the Null Hypothesis.

There is enough evidence to support the alternative hypothesis.

-
- If

$$p > \alpha$$

Fail to reject the Null Hypothesis.

There isn't enough evidence to say the claim is false.

Notice we say **"fail to reject"**, not **"accept"** the null hypothesis. There may simply not be enough evidence.

Example

Suppose

$$\alpha = 0.05$$

Case 1

$$p = 0.02$$

Since

$$0.02 < 0.05$$

Reject H_0 .

Case 2

$$p = 0.18$$

Since

$$0.18 > 0.05$$

Fail to reject H_0 .

d) Z-test

A **Z-test** is used to determine whether the sample mean is significantly different from the population mean **when the population standard deviation is known** or the sample size is large (typically ($n \geq 30$)).

The test statistic is:

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

where:

- $\bar{x}$ = sample mean
- μ = population mean under H_0
- σ = population standard deviation
- n = sample size

A larger absolute Z-value means the sample mean is farther from the hypothesized mean.

Z-Test ($\alpha = 0.05$)

Test Type	Critical Value
One-tailed	$\pm 1.645^*$
Two-tailed	± 1.96

*More precisely:

- Right-tailed: +1.645
- Left-tailed: -1.645

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t-test

A **t-test** is similar to a Z-test but is used when:

- The population standard deviation is **unknown**.
- The sample size is **small** (typically ($n < 30$)).

The test statistic is:

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

where:

- s = sample standard deviation

The t-distribution looks like the normal distribution but has **heavier tails**, reflecting greater uncertainty from estimating the standard deviation.

Types of t-tests

1. **One-sample t-test** – Compare one sample mean with a known value.
2. **Independent t-test** – Compare the means of two independent groups.
3. **Paired t-test** – Compare measurements before and after treatment on the same subjects.

T-Test ($\alpha = 0.05$)

Sample Size (n)	df	Critical t-value (Two-tailed)
5	4	2.776
10	9	2.262
20	19	2.093

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Difference Between Z-test and t-test

Z-test

Population standard deviation known

Usually large sample

Uses Normal (Z) distribution

t-test

Population standard deviation unknown

Usually small sample

Uses Student's t-distribution

e) Chi-square Test

The **Chi-square** (χ^2) test is used for **categorical data** rather than numerical data.

It checks whether the observed frequencies differ significantly from the expected frequencies.

The statistic is:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

where:

- O = Observed frequency
- E = Expected frequency

Example

Suppose a fair die should produce each number about 100 times in 600 rolls.

If the observed counts are very different from the expected counts, the Chi-square test helps determine whether the difference is due to chance or if the die may be biased.

Uses

- Goodness-of-fit test
 - Test of independence (e.g., "Is gender related to product preference?")
 - Test of homogeneity
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f) ANOVA (Analysis of Variance)

Suppose you want to compare the average exam scores of **three teaching methods**.

You could perform multiple t-tests, but doing many t-tests increases the probability of making a Type I Error.

Instead, we use **ANOVA**.

ANOVA tests whether **at least one group mean differs from the others**.

It does **not** tell you which group is different; it only tells you that a difference exists.

The ANOVA test statistic is:

$$F = \frac{\text{Between-group variation}}{\text{Within-group variation}}$$

Interpretation:

- Large F : Group means are likely different.
- Small F : Differences are probably due to random variation.

Example

Compare average crop yields using four different fertilizers.

ANOVA determines whether the fertilizers produce significantly different average yields.

g) Applications of Hypothesis Testing in Machine Learning

Hypothesis testing is widely used in ML because we often need to decide whether observed differences are real or just due to randomness.

1. Feature Selection

Determine whether a feature has a statistically significant relationship with the target variable.

Example:

Is **Age** significantly related to whether a customer buys insurance?

2. A/B Testing

Compare two versions of a website, advertisement, or recommendation algorithm.

Example:

Does the new homepage increase user clicks?

3. Comparing ML Models

Determine whether Model A is actually better than Model B or if the observed accuracy difference occurred by chance.

4. Feature Engineering

Check whether adding a new feature significantly improves model performance.

5. Medical AI

Determine whether a new AI-assisted diagnostic system performs significantly better than an existing method.

6. Data Validation

Check whether data collected from different sources comes from the same distribution or whether significant changes have occurred.

Practical Applications of Statistics in ML

These three topics explain how statistics is actually used when building and comparing machine learning models.

a) Feature Selection Using Statistical Tests

A machine learning dataset can contain many features, but not all of them are useful.

Feature selection means identifying useful features and removing irrelevant or redundant ones.

Statistical tests help determine whether a feature has a meaningful relationship with the target.

For example, suppose you're predicting whether someone buys a product:

- Age

- Salary
- Height
- Favorite color
- Experience

You can use statistical tests to determine which features are associated with Purchased.

Common tests include:

- t-test → numerical feature vs. two-class categorical target
- ANOVA → numerical feature vs. multiple classes
- Chi-square test → categorical feature vs. categorical target
- Correlation → relationship between numerical variables

The p-value is often used to judge statistical evidence:

$$p < 0.05$$

generally indicates statistically significant evidence of a relationship.

But remember: statistical significance doesn't automatically mean the feature will improve the ML model. You should ultimately verify feature usefulness using model performance.

b) Model Evaluation Metrics

After training a model, we need to determine:

"How well does it actually perform?"

The metric depends on whether you're solving classification or regression.

For classification, common metrics are:

Accuracy

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Measures the proportion of predictions that are correct.

Precision

$$\text{Precision} = TP / (TP + FP)$$

Of everything predicted as positive, how much was actually positive?

Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Of all actual positive cases, how many did the model find?

F1 Score

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Balances precision and recall.

ROC-AUC measures how well the model separates positive and negative classes across different classification thresholds.

For regression, common metrics include:

- MAE → average absolute error
- MSE → average squared error
- RMSE → square root of MSE
- R^2 → how well the model explains variation in the target

The important point is that there is no single best metric for every ML problem.

For example, in disease detection, Recall may be more important because missing a sick patient can be serious. In spam detection, Precision may be more important because incorrectly marking genuine emails as spam is undesirable.

c) Statistical Significance in Model Comparison

Suppose you train two models:

Model A → 90% accuracy

Model B → 92% accuracy

It looks like Model B is better.

But we need to ask:

"Is that 2% improvement actually meaningful, or could it have happened because of random variation in the data?"

This is where statistical significance comes in.

You can evaluate both models multiple times, commonly using cross-validation:

Model A Model B

Fold 1 90% 92%

Fold 2 91% 93%

Fold 3 89% 92%

Fold 4 90% 94%

Fold 5 91% 93%

Then a statistical test can be used to compare the paired results.

For example:

- Paired t-test
- Wilcoxon signed-rank test
- McNemar's test for certain paired classification predictions

The hypotheses could be:

$H_0: \mu_A = \mu_B$

$H_1: \mu_A \neq \mu_B$

If:

$p < 0.05$

we have evidence that the models' performances differ significantly.

If:

$p > 0.05$

we don't have enough evidence to conclude that the difference is statistically significant.

One Important Distinction

Statistical significance and practical significance aren't the same.

Suppose:

Model A = 90.00%

Model B = 90.01%

The difference could be statistically significant with a huge dataset, but a 0.01% improvement may not be practically useful.

So when choosing a model, you consider both:

"Is the improvement statistically real?"

and

"Is the improvement actually useful?"

In One Line

Feature selection → Which features should I use?

Model evaluation → How well does my model perform?

Statistical significance → Is one model's better performance actually meaningful or could it be random?